



DATA ANALYSIS FINAL PROJECT

Startup Ecosystem Analysis

Data-Driven Insights for Investment Decision Making

Crunchbase 2013 Snapshot · 11 Tables · 1M+ Records



Python



SQL Server



Power BI

Project Roadmap

End-to-end data analysis pipeline across 6 phases



01 Python

Data loading, cleaning & EDA across 11 tables using Pandas



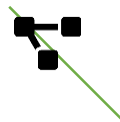
02 SQL Server

Schema design, ERD, FK constraints & analytical queries



03 Power Query

Data transformation, type corrections & column extraction



04 Data Modeling

Star schema, relationships & DAX measures in Power BI



05 Dashboard

5-page interactive dashboard with KPIs, charts & slicers



06 GitHub

Structured repository with README, SQL, notebooks & .pbix

Crunchbase 2013 — Dataset Overview

11 interrelated tables covering the global startup ecosystem up to December 2013

**196,553**

Companies

**52,680**

Funding Rounds

**80,902**

Investments

**9,561**

Acquisitions

**1,254**

IPOs

**\$419.7B**

Total Raised

**\$8.0M**

Avg per Round

**175**

Countries

**10,727**

Cities

**Venture**

Top Round Type



Strategic Question: What factors determine a startup's ability to raise higher funding or reach Acquisition / IPO?

Data Cleaning & EDA

Pandas pipeline applied across all 11 tables

Cleaning Decision Framework

- ✓ **Dates:** Never imputed — left as NaT (logical nulls preserved)
- ✓ **Foreign Keys:** Rows missing FK are dropped — cannot be joined
- ✓ **Logical Nulls:** closed_at = null means still open. Kept as-is
- ✓ **> 75% Missing:** Column dropped — adds no analytical value
- ✓ **Categorical:** Filled with Mode or 'Unknown' (e.g. category_code)
- ✓ **Numeric:** Median used (resistant to outliers like \$2.6T deal)
- ✓ **0 Values:** funding_total_usd null → 0 (no funding received)

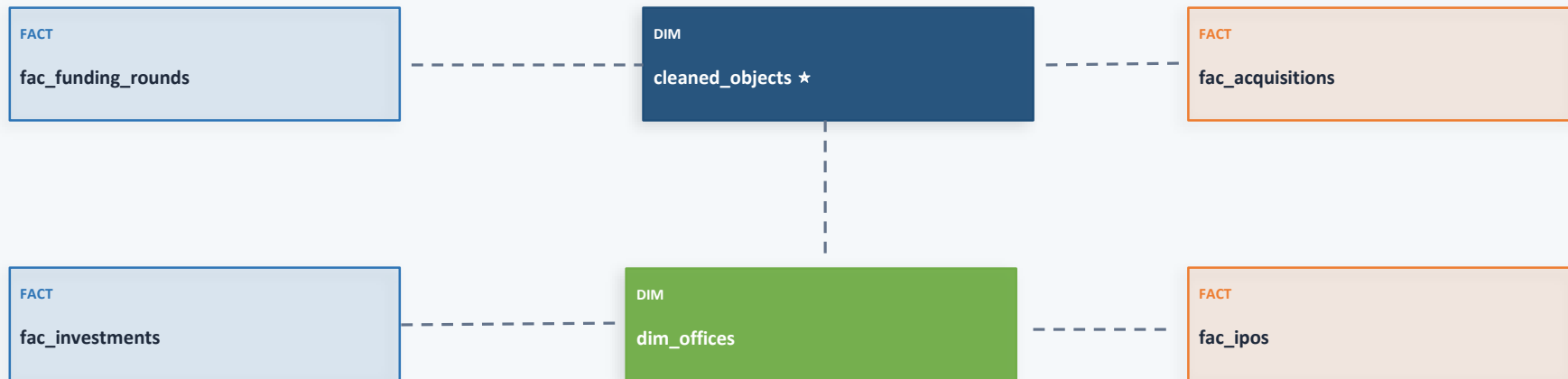
Post-Cleaning Summary

acquisitions	9,561 rows	9 cols	1 FK row dropped
funding_rounds	52,680 rows	18 cols	248 rows (no date)
investments	80,902 rows	6 cols	No changes needed
ipos	1,254 rows	11 cols	5 FK rows dropped
objects	462,651 rows	28 cols	12 columns dropped
offices	112,718 rows	11 cols	(0,0) coords filtered

✓ Zero duplicate rows found across all 11 tables

Database Design & Schema

Relational model with FK constraints and analytical queries



SQL Pipeline Steps:

- ✓ CREATE DATABASE companies_startups
- ✓ Orphan records deleted (all FK tables)
- ✓ UNIQUE constraint on funding_round_id
- ✓ FK constraints established (15 relationships)
- ✓ Self-join on cleaned_objects.parent_id
- ✓ Analytical SELECT queries written for Power BI

Data Model & DAX Measures

Star schema with 7 tables and 15+ calculated measures

Star Schema — Power BI Model

	dim_companies	196K companies — central dimension
	dim_investors	Filtered from objects (investment_rounds > 0)
	dim_date	Calendar table 1990–2013 with Year/Month/Quarter
	dim_offices	Geographic data with lat/long
	fac_funding_rounds	52K rounds — primary fact table
	fac_investments	80K investment records
	fac_acquisitions	9.5K acquisition deals
	fac_ipos	1.2K IPO records

Key DAX Measures

Total Raised USD	SUM of raised_amount_usd
Avg / Median Raised	AVERAGE & MEDIAN per round
Success Rate %	DIVIDE(Acquired + IPO, Total)
YoY Funding Growth	DATEADD comparison year-over-year
Avg Days to First Funding	AVERAGEX with DATEDIFF
Investments by Investor	CALCULATE + USERELATIONSHIP
% of Total Funding	DIVIDE with ALL() for total context
Round Size Category	SWITCH on raised_amount_usd ranges

Interactive Dashboard — 5 Pages

Decision-support tool with KPIs, slicers and drill-through visuals



Page 1 · Overview

KPI cards · Donut · Line trend · Top categories · Funded % rate



Page 2 · Funding Analysis

Avg by round type · Combo chart · Treemap · YoY growth card



Page 3 · Startup Journey

Success rate · Acquisitions by year · IPOs · Scatter plot



Page 4 · Investor Analysis

Top 20 investors · Activity trend · Sector treemap



Page 5 · Geographic

Filled map · Top countries · Top cities · Avg funding comparison

Primary:

#1F4E79

Accent:

#ED7D31

Font:

Segoe UI

KPI:

28pt Bold

Cards:

200×100px

Key Findings Across 4 Analytical Axes

Descriptive + Diagnostic analysis of the 2005–2013 startup ecosystem

Funding Analysis

1. \$419.7B total raised across 52,680 rounds
2. Venture is the dominant round type (29%)
3. Median \$1.6M vs Mean \$8.0M — extreme outliers present
4. Consistent YoY growth from 2005 to 2013
5. Private equity rounds have the highest average ticket size

Startup Journey

1. 93.5% of companies are still operating
2. Success rate (IPO + Acquired): ~5.4%
3. Software is the top sector by company count
4. Acquisitions peaked in 2011–2012
5. Companies with more rounds show higher survival rates

Geographic Analysis

1. 80,902 investment records from 17K+ unique investors
2. Top investors concentrated in Software & Mobile
3. Investment activity grew steadily post-2008
4. Most active investors are VC funds, not individuals

1. USA dominates: 55K+ companies, 79% of total funding
2. Silicon Valley (SF Bay) is the clear epicenter
3. UK, Israel & China are rising non-US hubs
4. Israel has highest avg funding per company

Repository Structure

crunchbase-startup-ecosystem-analysis



data/raw/

11 original CSV files from Kaggle



data/cleaned/

6 cleaned tables exported from Python



notebooks/

01_data_cleaning_eda.ipynb



sql/

01_setup · 02_cleaning · 03_queries



powerbi/

startup_ecosystem.pbix



docs/

README_EN.docx · schema.png · dashboard_preview.png



README.md includes: Project overview · Business Questions · Tools · File structure · EDA summary · Dashboard screenshots



Thank You

Startup Ecosystem Analysis — Crunchbase 2013

Python · Data Cleaning & EDA

SQL Server · Schema, FK Design & Select Query

Power BI · Power Query + Model + DAX + Dashboard

GitHub · Documented Repository